CSA1704-ARTIFICAL INTELLIGENCE LAB

4.write a python program for cript arithemantic problem

Aim:

To write a Python program to solve a cryptarithmetic problem by assigning unique digits to letters.

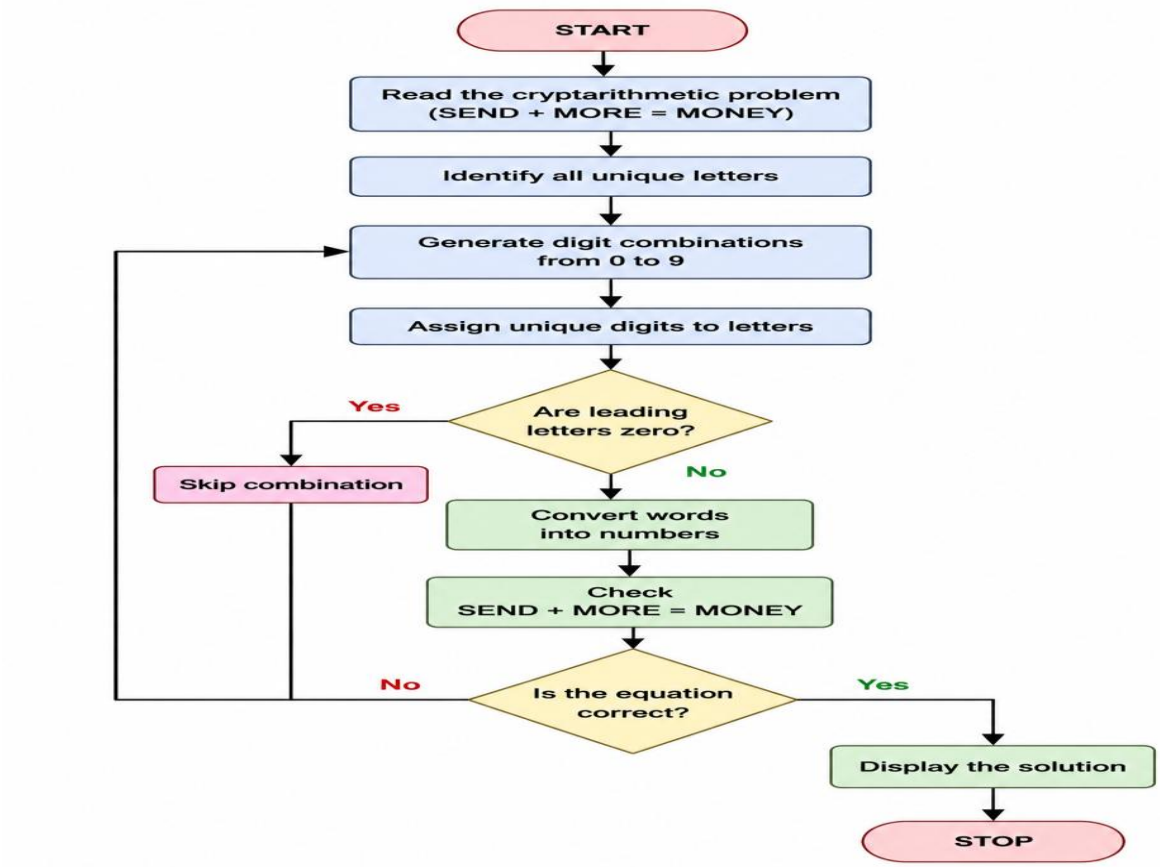
Objective:

To find a valid digit assignment that satisfies the given arithmetic equation while ensuring each letter represents a unique digit.

Algorithm:

1. Start the program.
2. Import the permutations function from the itertools module.
3. Store all unique letters of SEND + MORE = MONEY.
4. Generate different digit combinations from 0 to 9.
5. Assign one unique digit to each letter.
6. Ensure that S and M are not assigned zero because they are leading letters.
7. Convert the words SEND, MORE, and MONEY into numbers.
8. Check whether SEND + MORE = MONEY.
9. Display the valid solution.
10. Stop the program.

Flowchart:



Code:

```
from itertools import permutations

letters = "SENDMORY"

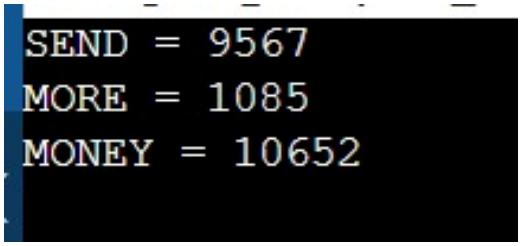
for p in permutations("0123456789", len(letters)):
    d = dict(zip(letters, map(int, p)))

    if d['S'] == 0 or d['M'] == 0:
        continue

    send = 1000*d['S'] + 100*d['E'] + 10*d['N'] + d['D']
    more = 1000*d['M'] + 100*d['O'] + 10*d['R'] + d['E']
    money = 10000*d['M'] + 1000*d['O'] + 100*d['N'] + 10*d['E'] + d['Y']
```

```
    if send + more == money:
        print("SEND =", send)
        print("MORE =", more)
        print("MONEY =", money)
        break
```

Output:



Result:

The Python program successfully solved the cryptarithmetic problem and displayed the correct digit assignment

Conclusion:

Thus, the Python program successfully solved the cryptarithmetic problem by assigning unique digits to each letter and verifying the given arithmetic equation.